Calculus and Linear Algebra

The Derivative: Definition, Meaning and First Rules

Lecture 17

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Lecture outline

- Average rate of change and secant lines.
- Tangent line as a limiting position of secants.
- Rigorous definition of derivative.
- Derivatives from the definition.
- One-sided derivatives and non-differentiability.
- Differentiability implies continuity.
- Constant, sum, product, quotient and power rules.
- Tangent lines and marginal quantities in economics.

Main goal

Understand the derivative first as a **limit and a rate of change**, and only then as a computational tool.

Average rate of change

For two points x_0 and $x_0 + h$, the average rate of change of f is

$$\boxed{\frac{f(x_0 + h) - f(x_0)}{h}}, \qquad h \neq 0.$$

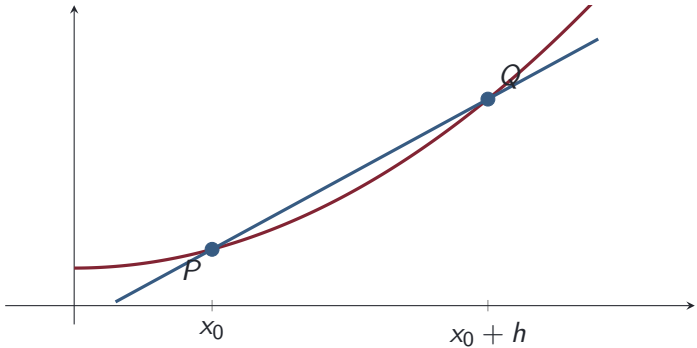
Geometrically, this is the slope of the secant line through

$$P = (x_0, f(x_0)), \qquad Q = (x_0 + h, f(x_0 + h)).$$

Interpretation

It measures the change in output per unit change in input over a finite interval.

Secant line: the geometric picture



For a finite h, the secant slope describes change across an interval, not at a single point.

From secants to the tangent

Move Q toward P by letting

$$h \to 0.$$

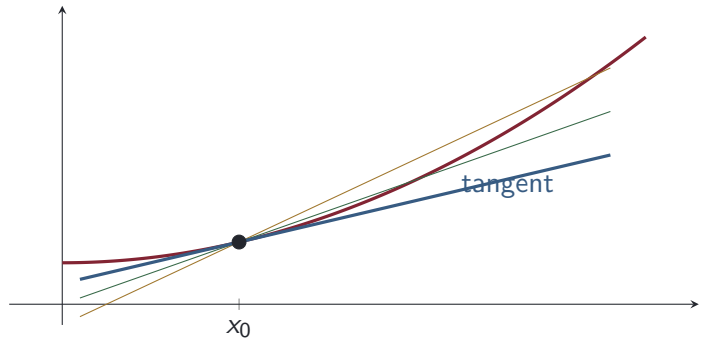
If the secant slopes approach a finite limit, that limit is the slope of the tangent line at x_0.

$$\frac{f(x_0 + h) - f(x_0)}{h} \quad \xrightarrow[h \to 0]{} \quad \text{slope of the tangent.}$$

The key idea

The derivative is an **instantaneous** rate of change obtained as a limit of **average** rates of change.

Secants approaching the tangent



The limiting line has slope $f'(x_0)$.

Derivative at a point: definition

Definition Let f be defined near x_0. If the following finite limit exists,

$$\boxed{f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h},}$$

then f is **differentiable at** x_0. The quotient

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

is called the **difference quotient**.

Equivalent form of the definition

Set

$$x = x_0 + h.$$

Then $h \to 0$ is equivalent to $x \to x_0$, so

$$\boxed{f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.}$$

Both definitions describe exactly the same limit.

Which form should we use?

The h-form is often convenient for calculation; the $x \to x_0$ form makes the geometric meaning especially transparent.

Notation for derivatives

Several notations are standard:

$$\boxed{f'(x)} \qquad \boxed{\frac{df}{dx}} \qquad \boxed{\frac{dy}{dx}} \qquad \boxed{Df(x)}.$$

At a particular point x_0:

$$f'(x_0).$$

Interpretation

The derivative is itself a function: it assigns to each differentiability point the slope of the tangent there.

Derivative from the definition: $f(x) = x^2$

Let

$$f(x) = x^2.$$

Then

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h}.$$

Expand:

$$\frac{x^2 + 2xh + h^2 - x^2}{h} = 2x + h.$$

Taking the limit,

$$\boxed{f'(x) = \lim_{h \to 0} (2x + h) = 2x.}$$

This result comes directly from the definition, not from a memorized rule.

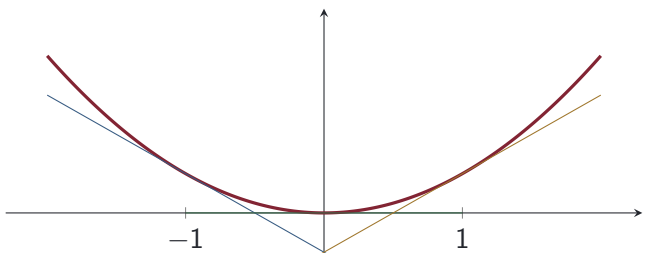
What does $f'(x) = 2x$ mean?

For the parabola

$$f(x) = x^2,$$

the tangent slope depends on the point:

$$f'(-1) = -2, \qquad f'(0) = 0, \qquad f'(1) = 2.$$



Negative slope, zero slope and positive slope correspond to different local behaviours.

Derivative from the definition: $f(x) = x^3$

$$\frac{(x+h)^3 - x^3}{h} = \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$
$$= 3x^2 + 3xh + h^2.$$

Therefore

$$\boxed{\frac{d}{dx}x^3 = 3x^2.}$$

At $x = 0$, the tangent is horizontal even though the function continues to increase through the origin.

The derivative of a constant

Let

$$f(x) = c.$$

Then

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0.$$

Hence

$$\boxed{f'(x) = 0.}$$

Geometrically, the graph is a horizontal line, whose slope is zero everywhere.

One-sided derivatives

At some points, the behaviour from the left and right may differ.

$$f'_-(x_0) = \lim_{h \to 0^-} \frac{f(x_0 + h) - f(x_0)}{h},$$

$$f'_+(x_0) = \lim_{h \to 0^+} \frac{f(x_0 + h) - f(x_0)}{h}.$$

The ordinary derivative exists iff

$$\boxed{f'_-(x_0) = f'_+(x_0)}$$

and both are finite.

A corner: $f(x) = |x|$

At $x_0 = 0$,

$$\frac{|h| - 0}{h} = \frac{|h|}{h}.$$

If $h > 0$,

$$\frac{|h|}{h} = 1,$$

while if $h < 0$,

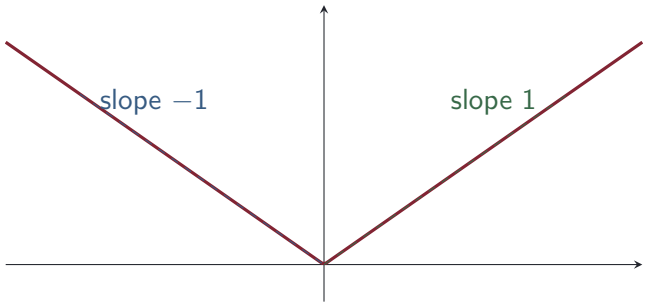
$$\frac{|h|}{h} = -1.$$

Thus

$$\boxed{f'_-(0) = -1 \neq 1 = f'_+(0),}$$

so $|x|$ is not differentiable at 0.

The corner is visible in the graph



The graph is continuous, but there is no unique tangent slope at the corner.

How differentiability can fail

A function may fail to be differentiable because of:

- a **corner**, such as $|x|$ at 0;
- a **cusp**;
- a **vertical tangent**, where the slope becomes infinite;
- a **discontinuity**.

Necessary condition

Every differentiable function is continuous. Therefore any discontinuity automatically prevents differentiability.

Differentiability implies continuity

Theorem If f is differentiable at x_0, then f is continuous at x_0.

$$\boxed{f \text{ differentiable at } x_0 \implies f \text{ continuous at } x_0.}$$

The converse is false. Example:

$$f(x) = |x|$$

is continuous at 0 but not differentiable there.

Proof: differentiability implies continuity

For $x \neq x_0$, write

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0}(x - x_0).$$

If f is differentiable at x_0, then as $x \to x_0$,

$$\frac{f(x) - f(x_0)}{x - x_0} \to f'(x_0), \qquad x - x_0 \to 0.$$

Therefore

$$f(x) - f(x_0) \to f'(x_0) \cdot 0 = 0.$$

Hence

$$\boxed{\lim_{x \to x_0} f(x) = f(x_0).}$$

So f is continuous at x_0.

The derivative as a function

If f is differentiable at every point of an interval, we obtain a new function

$$f' : x \mapsto f'(x).$$

Example:

$$f(x) = x^2 \qquad \Longrightarrow \qquad f'(x) = 2x.$$

The original function gives **levels**; the derivative gives **local rates of change**. Later we will use the sign of f' to study increasing and decreasing behaviour.

Basic differentiation rules

Once the derivative is understood as a limit, we can use rules derived from that definition. If c is constant:

$$\boxed{(cf)' = cf'.}$$

If f and g are differentiable:

$$\boxed{(f+g)' = f' + g',}$$

$$\boxed{(f-g)' = f' - g'.}$$

These rules make differentiation linear.

Power rule

For positive integers n,

$$\boxed{\frac{d}{dx}x^n = nx^{n-1}.}$$

Examples:

$$\frac{d}{dx}x^4 = 4x^3, \qquad \frac{d}{dx}x^7 = 7x^6.$$

Combined with linearity:

$$\frac{d}{dx}(3x^4 - 2x^2 + 5x - 7) = 12x^3 - 4x + 5.$$

Why the power rule works

From the difference quotient,

$$\frac{(x+h)^n - x^n}{h}.$$

Using the binomial expansion,

$$(x+h)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} h^j = x^n + nx^{n-1}h + \binom{n}{2} x^{n-2}h^2 + \cdots + h^n.$$

After subtracting x^n and dividing by h:

$$nx^{n-1} + \binom{n}{2} x^{n-2}h + \cdots + h^{n-1}.$$

Letting $h \to 0$ leaves only

$$\boxed{nx^{n-1}.}$$

Product rule

For differentiable functions f and g,

$$\boxed{(fg)' = f'g + fg'.}$$

Important

In general,

$$(fg)' \neq f'g'.$$

Example:

$$f(x) = x^2, \qquad g(x) = x + 1.$$

Then

$$(fg)' = 2x(x+1) + x^2 = 3x^2 + 2x.$$

Proof of the product rule

Start from

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

Add and subtract $f(x+h)g(x)$:

$$\frac{f(x+h)[g(x+h) - g(x)]}{h} + \frac{g(x)[f(x+h) - f(x)]}{h}.$$

As $h \to 0$, differentiability implies continuity, so

$$f(x+h) \to f(x).$$

Therefore

$$\boxed{(fg)'(x) = f(x)g'(x) + g(x)f'(x).}$$

Quotient rule

If $g(x) \neq 0$, then

$$\boxed{\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.}$$

Example:

$$f(x) = \frac{x^2 + 1}{x}.$$

Then

$$f'(x) = \frac{(2x)x - (x^2 + 1)(1)}{x^2} = \frac{x^2 - 1}{x^2}, \qquad x \neq 0.$$

Example: derivative of a polynomial

Differentiate

$$f(x) = 4x^5 - 3x^3 + 2x - 8.$$

Using the power rule term by term:

$$\boxed{f'(x) = 20x^4 - 9x^2 + 2.}$$

No limit calculation is needed once the basic rules have been established.

But remember

The differentiation rules are consequences of the limit definition; they do not replace its conceptual meaning.

Example: product and quotient together

Differentiate

$$f(x) = \frac{x^2(x+1)}{x-1}, \qquad x \neq 1.$$

Let

$$N(x) = x^2(x+1), \qquad D(x) = x - 1.$$

Then

$$N'(x) = 2x(x+1) + x^2 = 3x^2 + 2x, \qquad D'(x) = 1.$$

Hence

$$f'(x) = \frac{(3x^2 + 2x)(x-1) - x^2(x+1)}{(x-1)^2}.$$

The expression may be simplified after differentiation.

Equation of the tangent line

If f is differentiable at x_0, the tangent line has slope

$$m = f'(x_0)$$

and passes through

$$(x_0, f(x_0)).$$

Therefore

$$\boxed{y - f(x_0) = f'(x_0)(x - x_0).}$$

Equivalently,

$$\boxed{y = f(x_0) + f'(x_0)(x - x_0).}$$

This formula is also the first local linear approximation to f.

Tangent line example

Find the tangent to

$$f(x) = x^2$$

at $x_0 = 1$.

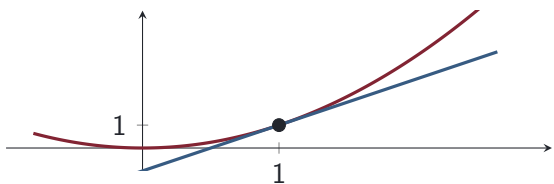
$$f(1) = 1, \qquad f'(x) = 2x, \qquad f'(1) = 2.$$

Thus

$$y - 1 = 2(x - 1),$$

so

$$\boxed{y = 2x - 1.}$$



Exercise: tangent from first principles

Exercise For

$$f(x) = x^2 + 1,$$

find the tangent line at $x_0 = 2$. Suggested steps:

1. compute $f(2)$;
2. use the derivative $f'(x) = 2x$;
3. evaluate $f'(2)$;
4. use the point–slope equation.

Exercise: solution

Solution

$$f(2) = 5, \qquad f'(2) = 4.$$

Therefore

$$y - 5 = 4(x - 2),$$

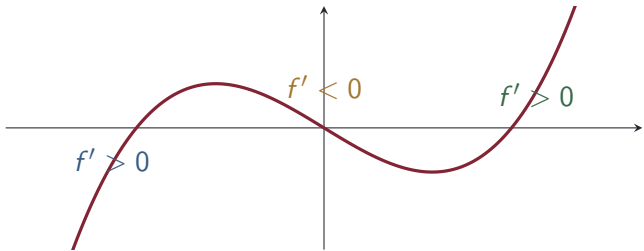
so

$$\boxed{y = 4x - 3.}$$

The tangent is the best first-order linear description of the graph near $x = 2$.

Derivative and local direction

If the tangent slope is positive at a point, the graph is locally rising there. If it is negative, the graph is locally falling there.



In later lectures we will turn this intuition into a systematic monotonicity test.

Exercise: differentiate

Exercise Find the derivative of each function.

1. $f(x) = 5x^4 - 2x^2 + 3$.
2. $g(x) = (x^2 + 1)(x - 3)$.
3. $h(x) = \dfrac{x^2 + 2x}{x + 1}$, $x \neq -1$.
4. $p(x) = 7$.

Identify which rule is used at each step.

Exercise: solutions

Solution

$$f'(x) = 20x^3 - 4x.$$

$$g'(x) = 2x(x-3) + (x^2+1) = 3x^2 - 6x + 1.$$

$$h'(x) = \frac{(2x+2)(x+1) - (x^2+2x)}{(x+1)^2}.$$

After simplification,

$$h'(x) = \frac{x^2+2x+2}{(x+1)^2}.$$

Finally,

$$p'(x) = 0.$$

Concept check: true or false?

Exercise Decide whether each statement is true.

1. Every continuous function is differentiable.
2. Every differentiable function is continuous.
3. If $f'(x_0) = 0$, then the tangent is horizontal.
4. The derivative is an average rate of change over a finite interval.
5. A corner may occur at a point where the function is continuous.

Concept check: answers

Solution

1. False: $|x|$ is continuous but not differentiable at 0.
2. True.
3. True.
4. False: it is the limit of average rates, hence an instantaneous rate.
5. True: again $|x|$ at 0.

What we know now

- The derivative is defined by a limit of difference quotients.
- Geometrically, it is the tangent slope.
- Economically, it measures a marginal or local rate of change.
- One-sided derivatives detect corners and other failures of differentiability.
- Differentiability implies continuity.
- Basic algebraic rules make derivatives efficient to compute.

The central formula

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

Looking ahead

The next step is to enlarge our differentiation toolbox:

- derivatives of exponential and logarithmic functions;
- derivatives of trigonometric functions;
- the **chain rule**;
- derivatives of composite functions;
- increasingly complex exercises.

Then we will use derivatives to study

monotonicity, extrema, concavity and complete graphs.